Triangle Books with a Page-Rooted Triangle–Leaf Branch

Abstract

Let B_m^{new} and B_m^{page} be obtained from a book of $m \geq 1$ triangles on a common spine by attaching a new triangle along the edge from one fixed spine endpoint to one exceptional page. Attach one leaf either to the new triangle vertex or to the exceptional page, respectively. We prove that every graphon of edge density $p \in [1/2, 1]$ satisfies, for either graph,

$$t(B_m^\star, W) \geq p^2(2p-1)^{m+1} = \Phi_{B_m^\star}(p).$$

The proof symmetrizes the exceptional branch over the book pages. Two fractional edge-rooted estimates follow from edge Jensen and the real-exponent weighted rooted-triangle inequality; one of them also uses a Hölder compression. The cases $m = 2$ prove NetworkX Atlas graphs 137 and 139.

1 The graph family and its target

Let $(\Omega, \mathcal{A}, \mu)$ be an arbitrary probability space and let $W\colon \Omega^2 \to [0,1]$ be symmetric and jointly measurable. Put

$$p = \int_{\Omega^2} W(x,y)\,\mathrm{d}\mu(x)\,\mathrm{d}\mu(y), \qquad d(x) = \int_\Omega W(x,y)\,\mathrm{d}\mu(y).$$

For $m \geq 1$, define two graphs as follows. Their spine vertices are a, b, their page vertices are $z_1, \ldots, z_m$, and their two additional vertices are u, v. Their common edges are

$$ab, \qquad az_i, bz_i \quad (1 \leq i \leq m), \qquad au, z_1u,$$

together with uv for B_m^{new} and with z_1v for B_m^{page}. Thus a, z_1, u is the added triangle; the superscript records whether v is a leaf at its new vertex u or at the old page vertex z_1.

After the ordered spine has been properly coloured, each book page and the new triangle vertex have $x-2$ choices, while the leaf has $x-1$ choices. Consequently

$$\chi_{B_m^\star}(x) = x(x-1)^2(x-2)^{m+1}, \qquad \chi(B_m^\star) = 3. \tag{1}$$

For $q = 1 - p > 0$, substitution in the chromatic target gives

$$\Phi_{B_m^\star}(p) = q^{m+4}\chi_{B_m^\star}(1/q) = p^2(2p-1)^{m+1}. \tag{2}$$

The right side is polynomial and supplies the continued value at $p = 1$.

2 Real weighted triangles

Define the rooted triangle density

$$\tau(x) = \int_{\Omega^2} W(x,y)W(x,z)W(y,z)\,\mathrm{d}\mu(y)\,\mathrm{d}\mu(z)$$

and, for every real $s \geq 0$,

$$M_s = \int_{\Omega} d(x)^s\,\mathrm{d}\mu(x),$$

with $d^0 = 1$ also on the zero-degree set.

Lemma 2.1 (Weighted rooted triangle)**.** *If $p > 0$, then for every real $h \geq 0$,*

$$\int_{\Omega} d(x)^h \tau(x)\,\mathrm{d}\mu(x) \geq \frac{2p-1}{p} M_{h+2}. \tag{3}$$

Proof. For a bounded kernel K, write $(T_K f)(x) = \int K(x,y)f(y)\,\mathrm{d}\mu(y)$ and use the usual L^2 inner product. Set

$$I_h = \int d^h \tau, \qquad J_h = \langle d^h, T_W d\rangle.$$

The inequality $rs \geq r + s - 1$ on $[0,1]^2$, applied to the two edges from the root of the triangle and multiplied by $d(x)^h W(y,z)$, yields

$$I_h \geq 2J_h - pM_h. \tag{4}$$

Put $U = 1 - W$, $u = T_U \mathbf{1} = 1 - d$, and $q = 1 - p$. Since

$$T_W d = d - q\mathbf{1} + T_U u,$$

we have

$$J_h = M_{h+1} - qM_h + \langle d^h, T_U u\rangle. \tag{5}$$

Symmetry of U gives

$$\begin{aligned} &2\left(\langle d^h, T_U u\rangle - \int d^h u^2\right) \\ &\quad = \int_{\Omega^2} U(x,y)\big(u(y) - u(x)\big)\big(d(x)^h - d(y)^h\big)\,\mathrm{d}\mu^2. \end{aligned}$$

The integrand is nonnegative because $d = 1 - u$ and $t \mapsto t^h$ is nondecreasing. Therefore

$$\langle d^h, T_U u\rangle \geq M_h - 2M_{h+1} + M_{h+2}.$$

Substitution into (5) and (4) gives

$$I_h \geq 2M_{h+2} - 2M_{h+1} + pM_h.$$

After subtracting the right side of (3), the remaining quantity is exactly

$$\frac{1}{p}\int_{\Omega} d(x)^h (d(x) - p)^2\,\mathrm{d}\mu(x) \geq 0.$$

No integrality of h was used. □

3 A fractional edge-rooted paw estimate

For $x, y \in \Omega$ and real $s \geq 0$, put

$$H_s(x,y) = \int_\Omega W(x,z)W(y,z)d(z)^s \,\mathrm{d}\mu(z).$$

Thus H_0 is the common-neighbour kernel, while H_1 is the rooted density obtained by putting one leaf at the new common neighbour.

Lemma 3.1 (Fractional edge-rooted paw)**.** *Let $p \in [1/2, 1]$ and $0 < \alpha \leq 1$. Then*

$$\int_{\Omega^2} W(x,y)H_0(x,y)H_1(x,y)^\alpha \,\mathrm{d}\mu^2 \geq \big(p(2p-1)\big)^{1+\alpha}. \tag{6}$$

Proof. Put $s = \alpha/(1+\alpha)$. Hölder's inequality applied to the measure $W(x,z)W(y,z)\,\mathrm{d}\mu(z)$ gives, pointwise in (x,y),

$$H_s \leq H_0^{1-s} H_1^s.$$

Since $(1-s)(1+\alpha) = 1$ and $s(1+\alpha) = \alpha$, this is equivalent to

$$H_0 H_1^\alpha \geq H_s^{1+\alpha}. \tag{7}$$

Because $p \geq 1/2 > 0$, the measure $W(x,y)\,\mathrm{d}\mu^2/p$ is a probability measure. Jensen's inequality and (7) imply

$$\begin{aligned} \int W H_0 H_1^\alpha &\geq \int W H_s^{1+\alpha} \\ &\geq p^{-\alpha} \left(\int W H_s \right)^{1+\alpha}. \end{aligned} \tag{8}$$

Fubini gives the exact identity

$$\int_{\Omega^2} W(x,y)H_s(x,y)\,\mathrm{d}\mu^2 = \int_\Omega d(z)^s \tau(z)\,\mathrm{d}\mu(z).$$

By Lemma 2.1 and Jensen for the convex power $t \mapsto t^{s+2}$,

$$\int W H_s \geq \frac{2p-1}{p} M_{s+2} \geq p^{s+1}(2p-1).$$

Insert this in (8). The choice of s gives

$$-\alpha + (s+1)(1+\alpha) = 1 + \alpha,$$

and hence the claimed lower bound. □

4 A fractional exceptional-page estimate

Lemma 4.1 (Fractional exceptional page)**.** *Let $p \in [1/2, 1]$ and $0 < \alpha \leq 1$. Then*

$$\int_{\Omega^2} W(x,y)d(y)^\alpha H_0(x,y)^{1+\alpha} \,\mathrm{d}\mu^2 \geq \big(p(2p-1)\big)^{1+\alpha}. \tag{9}$$

Proof. Put $s = \alpha/(1+\alpha)$. Since

$$d(y)^{\alpha} H_0(x,y)^{1+\alpha} = \big(d(y)^s H_0(x,y)\big)^{1+\alpha},$$

Jensen under the edge-biased probability measure gives

$$\int W d(y)^{\alpha} H_0^{1+\alpha} \geq p^{-\alpha} \left(\int W(x,y) d(y)^s H_0(x,y) \, \mathrm{d}\mu^2 \right)^{1+\alpha}.$$

Fubini gives

$$\int_{\Omega^2} W(x,y) d(y)^s H_0(x,y) \, \mathrm{d}\mu^2 = \int_{\Omega} d(y)^s \tau(y) \, \mathrm{d}\mu(y).$$

The final two lines of the proof of Lemma 3.1 now apply verbatim and give (9). □

5 The two branch orientations

Theorem 5.1 (Triangle books with one page-rooted triangle–leaf branch)**.** *For every integer $m \geq 1$, every graphon W of edge density $p \in [1/2, 1]$, and $\star \in \{\text{new}, \text{page}\}$,*

$$\boxed{t(B_m^{\star}, W) \geq p^2 (2p-1)^{m+1} = \Phi_{B_m^{\star}}(p).}$$

This is the complete interval required by $\chi(B_m^{\star}) = 3$.

Proof. First take B_m^{new}. Integrating the added leaf and then its neighbour gives the factor $H_1(x, z_1)$ on the exceptional book page. The m page variables form one orbit in the underlying book. Average the m equal integral representations obtained by moving the exceptional factor among the pages, and apply arithmetic–geometric mean pointwise. With x, y denoting the spine images, this gives

$$t(B_m^{\text{new}}, W) \geq \int_{\Omega^2} W(x,y) R(x,y)^m \, \mathrm{d}\mu^2, \tag{10}$$

where

$$R(x,y) = \int_{\Omega} W(x,z) W(y,z) H_1(x,z)^{1/m} \, \mathrm{d}\mu(z).$$

The pages factor independently after symmetrization, which justifies (10).

Jensen under the edge-biased probability measure gives

$$t(B_m^{\text{new}}, W) \geq p^{1-m} \left(\int_{\Omega^2} W(x,y) R(x,y) \, \mathrm{d}\mu^2 \right)^m. \tag{11}$$

Fubini and the definition of H_0 show that the inner integral is

$$\int_{\Omega^2} W(x,z) H_0(x,z) H_1(x,z)^{1/m} \, \mathrm{d}\mu(x) \, \mathrm{d}\mu(z).$$

Lemma 3.1 with $\alpha = 1/m$ bounds this by

$$\big(p(2p-1)\big)^{1+1/m}.$$

Substitution into (11) yields

$$t(B_m^{\text{new}}, W) \geq p^{1-m} \big(p(2p-1)\big)^{m+1} = p^2 (2p-1)^{m+1},$$

which is (2).

For B_m^{page}, integrating the leaf at z_1 and the new triangle vertex u gives the exceptional factor $d(z_1)H_0(x,z_1)$. The identical page-orbit symmetrization and outer edge Jensen give

$$t(B_m^{\mathrm{page}},W)\ge p^{1-m}\left(\int_{\Omega^2}W(x,y)S(x,y)\,\mathrm{d}\mu^2\right)^m,$$

where

$$S(x,y)=\int_\Omega W(x,z)W(y,z)\big(d(z)H_0(x,z)\big)^{1/m}\,\mathrm{d}\mu(z).$$

Fubini turns the integral in parentheses into

$$\int_{\Omega^2}W(x,z)d(z)^{1/m}H_0(x,z)^{1+1/m}\,\mathrm{d}\mu(x)\,\mathrm{d}\mu(z).$$

Lemma 4.1 with $\alpha=1/m$ bounds this by $\big(p(2p-1)\big)^{1+1/m}$. The same exponent calculation therefore proves the asserted bound for B_m^{page}. □

At $p=1/2$ the target is zero and all divisions above are only by the positive number p. At $p=1$, $W=1$ almost everywhere and both sides are one. For every balanced complete k-partite graphon, $k\ge 2$, one has $d=p$ and $H_s=p^s(2p-1)$ on every supported edge; hence all Hölder, Jensen, and symmetrization steps are equalities, as required.

6 Atlas 137, Atlas 139, and scope

Corollary 6.1 (Atlas 137 and 139)**.** *NetworkX Atlas graphs* 137 *and* 139*, with graph6 codes* `EjsG` *and* `Eju?`*, respectively, are positive. Each has order six, size eight, chromatic number three, and*

$$\chi_{F_{139}}(x)=x(x-1)^2(x-2)^3.$$

For $i\in\{137,139\}$, every graphon of edge density $p\in[1/2,1]$ satisfies

$$t(F_{139},W)\ge p^2(2p-1)^3=\Phi_{F_{139}}(p).$$

Proof. For Atlas 137, use spine 13, ordinary page 2, exceptional page 4, added triangle vertex 0, and leaf 5 at page 4. Its edge list is

$$01,04,12,13,14,23,34,45,$$

so it is B_2^{page}. For Atlas 139, use the same roles but put the leaf at the new vertex 0. Its edge list is

$$01,04,05,12,13,14,23,34,$$

so it is B_2^{new}. Theorem 5.1 applies to both. □

All integrands in the proof are bounded, nonnegative, and measurable, so Tonelli–Fubini applies directly on an arbitrary probability space. The argument concerns ordinary homomorphism densities; it uses no finite-step reduction, injective-copy interpretation, regularity, atomlessness, or minimum-degree hypothesis. Zero degrees cause no quotient.

The theorem requires one triangle–leaf branch rooted at a page and one fixed spine endpoint of a single triangle book, in either of the two stated leaf orientations. It does not assert arbitrary leaf, triangle-page, edge-gluing, or mixed-chordal closure. In particular, its proof uses both page-orbit symmetrization and the specific edge-rooted Hölder compression (7).

A What the Lean formalization actually proves

The Lean development formalizes the case $m = 2$, which is the only case the catalogue needs. The modules are `lean/Taeyoung/Methods/PagePawBranch/Core.lean` and `Rows.lean`; the two catalogue theorems are `satisfiesLowerBound_bookNew` (Atlas 139) and `satisfiesLowerBound_bookPage` (Atlas 137). The formalization departs from the argument above at two places, and this appendix says where.

A.1 The page-orbit symmetrization is a Hölder inequality

The proof of Theorem 5.1 obtains (10) by averaging m integral representations obtained by moving the exceptional factor among the pages, and applying arithmetic–geometric mean pointwise. Those m representations are indeed equal, but for a trivial reason: the m page factors are m factors of one product, so naming which of them carries the exceptional weight changes nothing, and the average is the same single quantity. Symmetrization therefore contributes no inequality, and the step from that product to $\iint W R^m$ is carried entirely by the pointwise bound $R^m \le G H_0^{m-1}$ below, which is Hölder, not arithmetic–geometric mean.

Peeling the leaf and the glued vertex gives the exact identity

$$t(B_m^{\mathrm{new}}, W) = \int_{\Omega^2} W(x,y)\, H_0(x,y)^{m-1}\, G(x,y)\, \mathrm{d}\mu^2, \qquad G(x,y) = \int_{\Omega} W(x,z) W(y,z) H_1(x,z)\, \mathrm{d}\mu(z), \tag{12}$$

with no inequality yet used. Hölder on the finite measure $W(x,z)W(y,z)\,\mathrm{d}\mu(z)$, applied to the pair of exponents m and $m/(m-1)$, gives

$$R(x,y) = \int_{\Omega} W(x,z) W(y,z) H_1(x,z)^{1/m}\, \mathrm{d}\mu(z) \le G(x,y)^{1/m} H_0(x,y)^{(m-1)/m}, \tag{13}$$

that is $R^m \le G H_0^{m-1}$, which together with (12) is exactly (10). At $m = 2$ the two exponents are both 2 and (13) is Cauchy–Schwarz. The same remark applies verbatim to B_m^{page}, with $H_1(x,z)$ replaced by $d(z)H_0(x,z)$.

The Lean proof therefore never symmetrizes. It works with the *branch operator*

$$\Lambda_h(x,y) = \int_{\Omega} W(x,z) W(y,z) h(x,z)\, \mathrm{d}\mu(z),$$

of which $H_0 = \Lambda_1$ is the special case $h \equiv 1$, proves (12) in the form

$$t = \int_{\Omega^2} W \cdot H_0 \cdot \Lambda_h, \qquad h = H_1 \;\; (B_2^{\mathrm{new}}), \qquad h(x,z) = d(z) H_0(x,z) \;\; (B_2^{\mathrm{page}}),$$

and replaces (13) by

$$\Lambda_{\sqrt{h}}(x,y)^2 \le H_0(x,y)\, \Lambda_h(x,y) \qquad (\texttt{sq_branchOp_le}),$$

one application of the weighted Cauchy–Schwarz inequality $(\int A\eta)^2 \le (\int A)(\int A\eta^2)$ with $A = W(x,\cdot)W(y,\cdot)$ and $\eta = \sqrt{h(x,\cdot)}$. This is the project's `PureChordal.WeightedCauchySchwarz`, already used for the page compression $H_{s/2}^2 \le H_0 H_s$.

A.2 Edge Jensen at exponent $3/2$, without convexity

Both fractional estimates of Sections 3 and 4 use Jensen for the convex power $t \mapsto t^{1+\alpha}$ under the edge-biased probability measure $W\,\mathrm{d}\mu^2/p$. At $\alpha = 1/2$ the exponent is $3/2$, and the development has no lemma for Jensen at a fractional power on a biased measure. It is not needed: writing

$$A = \int_{\Omega^2} WF, \qquad S = \int_{\Omega^2} W\sqrt{F}, \qquad T = \int_{\Omega^2} WF\sqrt{F}$$

for a measurable F with values in $[0,1]$, two applications of the same weighted Cauchy–Schwarz inequality — at weight W with $\eta = \sqrt{F}$, and at weight $W\sqrt{F}$ with the same η — give

$$S^2 \le pA, \qquad A^2 \le ST.$$

Eliminating S yields $A^4 \le S^2T^2 \le pAT^2$, hence

$$A^3 \le pT^2 \qquad (\texttt{cube_edge_le}), \tag{14}$$

which is the $\alpha = 1/2$ case of (8) after clearing denominators. The case $A = 0$ is separate and immediate. So the whole formalization uses one inequality, weighted Cauchy–Schwarz, at four different weights, and no genuine Hölder inequality and no convexity lemma anywhere.

A.3 The two comparison kernels

With (14) in hand, each row supplies a kernel F with two properties: the pointwise compression $F\sqrt{F} \le \sqrt{h}\,H_0$, and the scalar bound $\iint WF \ge p^{4/3}(2p-1)$. The generic conclusion

$$t \ge p^{-2}\Big(\int_{\Omega^2} WF\Big)^3 \ge p^2(2p-1)^3 \qquad (\texttt{branch_bound})$$

then holds for both rows at once. The two choices are:

- B_2^{new}: $F = H_{1/3}$. The compression squares to $H_{1/3}^3 \le H_0^2 H_1$, which is (7) at $\alpha = 1/2$ and is already in the development as `Link.cube_pageOp_le`, itself two staggered Cauchy–Schwarz steps. The scalar bound is $\iint WH_{1/3} = \int d^{1/3}\tau$ (`Link.integral_edge_pageOp`) together with Lemma 2.1 at $h = 1/3$ and Jensen at exponent $7/3$.
- B_2^{page}: $F(x,z) = d(z)^{1/3}H_0(x,z)$. The compression is an *identity*, both sides being $d(z)^{1/2}H_0(x,z)^{3/2}$. The scalar bound needs the Fubini identity

  $$\int_{\Omega^2} W(x,y)\,d(y)^s H_0(x,y)\,\mathrm{d}\mu^2 = \int_{\Omega} d(z)^s \tau(z)\,\mathrm{d}\mu(z),$$

  the counterpart of `integral_edge_pageOp` with the leaf weight carried by a spine endpoint rather than by the page vertex. It is proved as `integral_edge_degree_pageOp`.

In both cases the only fractional exponent surviving to the end is the $p^{4/3}$ of the scalar bound, and it is cubed back to p^4 exactly as in the three-page row `PageBook.Atlas138`.

A.4 Scope of the formalization

Only $m = 2$ is formalized. The general-m theorem is not: the two ingredients that would have to be generalized are (13) at m factors and (14) at exponent $1 + 1/m$, and both are stated in the Lean development at the square root only.

The Lean graphs use their own labelling, chosen so that peeling produces (12) directly: spine $0, 1$; exceptional page 2; ordinary page 3; glued vertex 4 on the page edge 0–2; leaf 5, at the glued vertex for B_2^{new} and at the exceptional page for B_2^{page}. The generated Atlas modules `Examples/Graph139.lean` and `Examples/Graph137.lean` carry the bound to the Atlas edge lists along an explicit relabelling whose adjacency correspondence is checked by kernel computation. The chromatic data (1) is proved, not assumed, by a three-step clique-attachment tower over K_3: pages attach to the 2-cliques $\{a, b\}$ and $\{a, z_1\}$, and the leaf to a single vertex, giving $x(x-1)^2(x-2)^3$ with chromatic number three.

Both catalogue theorems, and the twenty-one supporting lemmas, depend on exactly `[propext, Classical.choice, Quot.sound]`.